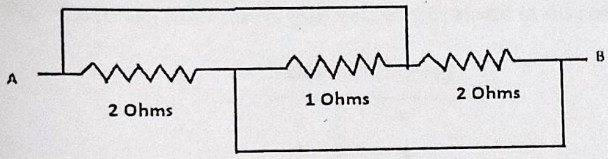
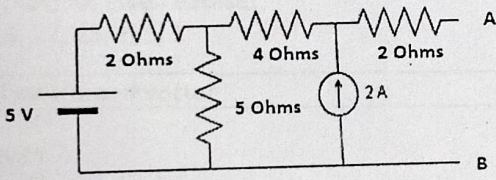
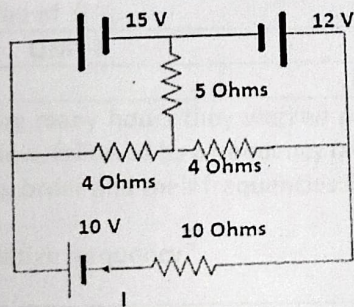


MID-TERM EXAMINATION
(Course Name: B. Tech:Reappear
(October, 2024) OFF LINE mode

Subject Code: BEC 110	Subject: Basic Electrical Engineering
Time: 1 ½ Hours	Maximum Marks: 30
Note: Q. 1 is compulsory.	

Q1	Q. 1 is compulsory.	(2.5*4)	
a)	Find the equivalent resistance between A and B. 	UNIT-1	CO ₁
b)	State and Prove Maximum Power Transfer Theorem. Derive the expression of Maximum Power transferred.	UNIT-1	
c)	Derive the expression of Power and Energy in an inductor.	UNIT-2	CO ₂
d)	Prove that the current in an R-C circuit leads the voltage.	UNIT-2	
UNIT-1 (CO₁)			(5,5)
Q2	(Attempt any Two Parts)		
a)	Find Thevenin's Equivalent circuit across AB. 		
b)	Determine the total current I in the given circuit. 		
c)	Derive the expression for conversion of a star load into delta load.		
UNIT-2 (CO₂)			(5,5)
Q3	(Attempt any Two Parts)		
a)	Find the average and RMS value of a sinusoidal ac waveform. Do find peak factor and form factor.		
b)	A pure resistance of 100 Ω is connected in series with a pure inductor of 50 mH. The series combination is connected across 240 V, 50 Hz supply. Find (i) Impedance (ii) Current (iii) Power Factor (iv) Voltage across R and C (v) Draw phasor diagram		
c)	Derive the expression of resonance frequency for a series RLC circuit, which is connected across an ac voltage source ($v = V_m \sin \omega t$). Also draw the phasor diagram and derive the following: (i) Total impedance (ii) Expression of current and (iii) Quality factor.		